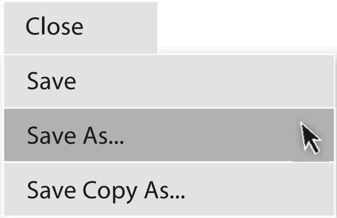
Drawing rectangles

Let's begin by importing the `turtle` module and using it to create a function that draws rectangles.

1 Create a new file

Open IDLE and create a new file. Save it as "robot_builder.py".



2 Import the Turtle module

Type this line at the top of your program. The command `import turtle as t` lets you use functions from the `turtle` module without having to type "turtle" in full each time. It's like calling someone whose name is Benjamin "Ben" for short.

```
import turtle as t
```

This gives the Turtle module the nickname "t".

3 Create a rectangle function

Now make the function to draw the blocks that you're going to use to build your robot. The function has three parameters: the length of the horizontal side; the length of the vertical side; and colour. You'll use a loop that draws one horizontal side and one vertical side each time it runs, and you'll make it run twice. Put this rectangle function under the code you added in Step 2.

Like all programming languages, Python uses the US spelling "color".

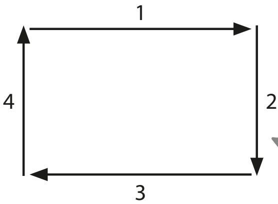
```
def rectangle(horizontal, vertical, color):
    t.pendown()
    t.pensize(1)
    t.color(color)
    t.begin_fill()
    for counter in range(1, 3):
        t.forward(horizontal)
        t.right(90)
        t.forward(vertical)
        t.right(90)
    t.end_fill()
    t.penup()
```

Put the turtle's pen down to start drawing.

Using `range(1, 3)` makes the loop run twice.

This block draws the rectangle.

Pull the turtle's pen back up to stop drawing.

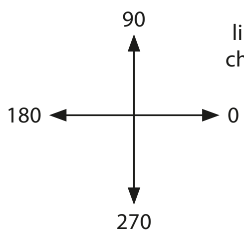


The turtle draws the sides in the order shown here.

EXPERT TIPS

Turtle mode

You'll be using the turtle in its standard mode. This means the turtle starts off facing the right side of the screen. If you set the heading (another word for direction) to 0, it will face right. Setting the heading to 90 makes it point to the top of the screen, 180 points it to the left, and 270 makes it point to the bottom of the screen.



The turtle normally looks like an arrowhead. This line changes it to a turtle shape.

```
t.shape('turtle')
t.setheading(0)
t.forward(80)
```